

TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (CHEMISTRY)

CHAPTER 5: CHEMICAL BONDING

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Chemistry)
Chapter Name:	Chemical Bonding

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TOPIC ROADMAP: VALENCE ELECTRONS AND BONDING TYPES



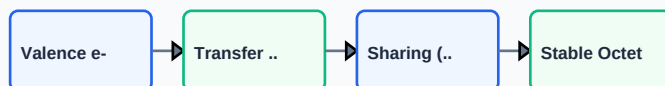
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Octet Rule	The tendency of atoms to prefer to have eight electrons in their valence shell by losing, gaining, or sharing electrons to achieve stability.
Ionic Bond	A chemical bond formed by the electrostatic attraction between oppositely charged ions, involving electron transfer (metals to non-metals).
Covalent Bond	A chemical bond formed by the sharing of one or more electron pairs between two non-metal atoms.
Coordinate Covalent Bond	A covalent bond where the shared electron pair is contributed by only one of the bonding atoms (donor-acceptor bond).
Lewis Dot Structure	A visual representation showing the valence shell electrons of an atom and how they are distributed in bonds.

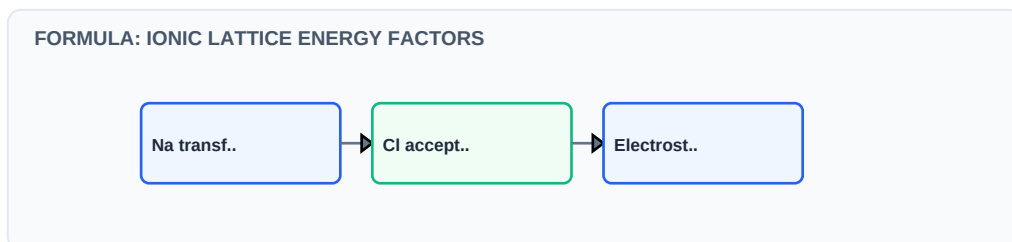
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

CHEMICAL OCTET BONDING FLOW



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Ionic bonding capacity = Number of electrons transferred to form stable ions.
- Covalent bonding capacity = Number of electron pairs shared between atoms.

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Describe the formation of Magnesium Chloride (MgCl₂) using ionic bonding.

Step-by-step Solution:

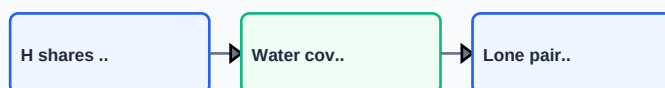
Magnesium (2,8,2) loses 2 valence electrons to become Mg²⁺. Two Chlorine atoms (2,8,7) each gain 1 electron to become Cl⁻. Electrostatic force binds them: Mg²⁺ + 2Cl⁻ → MgCl₂.

Example 2: Draw the covalent bonding sharing in a Methane (CH₄) molecule.

Step-by-step Solution:

Carbon (2,4) has 4 valence electrons and needs 4. Four Hydrogen atoms (1) each share 1 electron with Carbon, forming 4 single covalent C-H bonds.

WORKED: LEWIS STRUCTURE OF WATER MOLECULE



HOTS (Higher Order Thinking Skills) Challenge

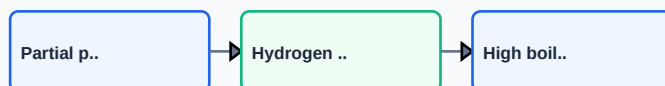
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Why do ionic compounds (like NaCl) have high melting points and conduct electricity in molten state, whereas covalent compounds (like glucose) do not?

Analytical Solution Strategy:

Ionic compounds form strong crystal lattices held by strong electrostatic forces, requiring high thermal energy to melt. In molten or aqueous states, these ions dissociate and are free to migrate, conducting current. Covalent molecules are bound by weak intermolecular forces, melting easily and containing no free ions or electrons to conduct electricity.

HOTS: POLARITY AND DIPOLE MOMENT INTERACTIONS



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Explain the conditions required for the formation of an ionic bond.

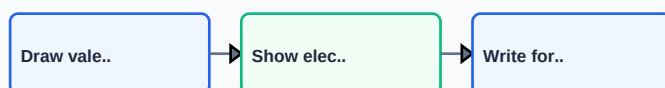
Q2 (2-Mark): Draw the Lewis dot structures of: H_2O , NH_3 , and CO_2 .

Q3 (2-Mark): What is a coordinate bond? Explain its formation in ammonium ion (NH_4^+).

Q4 (5-Mark): List four differences between ionic and covalent compounds.

Q5 (5-Mark): State the octet rule and discuss its limitations (e.g. SF_6 , PCl_5).

PRACTICE: DRAWING LEWIS DOT BOND DIAGRAMS



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Industrial Electrolyte Production

Ionic compounds are manufactured for batteries and physiological saline drinks to permit electrical conduction.

Application Case: Polymer Synthesis

Covalent materials (plastics, silicones) are synthesized to produce structural insulators and heat-resistant wraps.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: POLYMER CHEMISTRY CROSS-LINKING DESIGN



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm³, N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: CHEMICAL BONDING TYPES SUMMARY

